SYNTHETIC STUDY A | PASSAGE-GROUNDED ANSWERING

# Retrieval and hallucination in research Q&A

**Question:** Can retrieval-augmented generation reduce hallucination in research Q&A?

**Method:** Retrieve external knowledge passages before a language model drafts an answer.

**Finding:** Cited evidence helps readers check factual claims.

SYNTHETIC STUDY A | RETRIEVAL METHOD

# Dense similarity search for paper passages

**Method:** Dense vector similarity search ranks passages by semantic match to a question.

**Finding:** Retrieved context can improve factual accuracy over model weights alone.

**Limitation:** Different wording may hide a relevant result.

SYNTHETIC STUDY B | GRAPH-GUIDED RETRIEVAL

# Connecting concepts across research papers

**Question:** Can a knowledge graph surface evidence that dense similarity search misses?

**Method:** Link papers and concepts, then retrieve passages attached to nearby concepts.

**Finding:** Graph traversal can connect a topic to a method or result.

SYNTHETIC STUDY B | COMPARISON AND LIMITATIONS

# Hybrid retrieval and graph coverage

**Comparison:** Vector search finds semantic matches; graph traversal can boost linked chunks.

**Limitation:** Sparse or incorrect graph links can hide evidence. Thin graph coverage does not show field-wide absence.

**Open question:** How do graph upkeep costs compare with retrieval gains?

SYNTHETIC STUDY C | DOCUMENT INGESTION

# Chunking long papers into specific passages

**Question:** How should a paper be split into retrieval units?

**Method:** Extract page text, split it into short passages, and embed each passage.

**Limitation:** A chunk boundary can separate a claim from its caveat.

SYNTHETIC STUDY D | TRANSFORMER ARCHITECTURE

# Self attention and parallel sequence processing

**Method:** A transformer replaces recurrent computation with self attention over sequence positions.

**Finding:** Self attention allows parallel processing and faster training than recurrence.

**Limitation:** Long sequences use more memory.

SYNTHETIC STUDY D | ATTENTION MECHANISM

# Query, key, and value projections

**Method:** Tokens become query, key, and value vectors; query-key similarity sets weights over values.

**Finding:** Weighted values combine information across the sequence.

**Limitation:** Attention does not retrieve documents or prove an answer true.

SYNTHETIC STUDY E | KNOWLEDGE SOURCES

# Retrieval versus a parametric language model

**Comparison:** Retrieval systems consult an external corpus; a parametric language model answers from learned weights.

**Finding:** External passages can provide updateable evidence and citations.

**Limitation:** Answers depend on corpus coverage and ranking.

SYNTHETIC CROSS-PAPER COMPARISON

# Methods, evidence, and reported limitations

**Study A:** Uses passages to ground answers and reduce hallucination.

**Study B:** Combines dense similarity and graph traversal.

**Study C:** Studies chunking; passages may split a claim from its caveat.

SYNTHETIC OPEN-PROBLEM EXCERPT

# Evidence-backed gaps for a follow-up test

**Open gaps:** Test paraphrased citations, graph upkeep, retrieval latency, and chunk boundaries that split claims from limits.

**Evaluation:** Use these synthetic excerpts to practice comparison and retrieval evaluation.

**Scope:** All studies are invented, not published or field-wide evidence.